

Orlando Reyes

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Summary

Electronics engineering student specializing in embedded systems and low-level software. Hands-on experience with bare-metal STM32/ARM firmware, ESP32, FPGA/VHDL digital design, and Docker-based infrastructure. Comfortable working from register-level peripheral drivers up to Linux and Python automation tooling.

Education

Engineer in Electronics — Universidad Autónoma de Aguascalientes 08/2023 – Present (Expected 2027)
Vice Chair and founding member of the IEEE Student Branch at UAA. Technical documentation and publishing in LaTeX.

Technician in Mechatronics — CBTis No. 168 08/2019 – 07/2022
Hands-on training in diagnostics, maintenance of mechatronic equipment, AutoCAD drafting, and CNC programming.

Embedded & Software Projects

RISC-V Processor Core — Designed and implemented a custom 32-bit RISC-V processor core in VHDL, synthesized and executed on an FPGA.

IEEE-754 Floating-Point Calculator — Built a 32-bit IEEE-754 floating-point arithmetic unit in VHDL and ARM Assembly, with UART-based dynamic I/O.

STM32 Bare-Metal Drivers — Wrote register-level SPI drivers for ILI9341 and ST7735R TFT displays and integrated the XPT2046 resistive touch controller, working directly from datasheets.

SCADA System for Industrial PLC — Designed a low-cost SCADA system connecting a Linux-based industrial PLC to Node-RED for real-time HMI monitoring, with secure remote access over a WireGuard mesh VPN.

Custom PCB & Power Hardware Design — Designed minimum-system MCU development boards in KiCad, plus power electronics topologies with custom board layouts.

FPV Flight Controller Board Debugging — Assembled an FPV quadcopter running Betaflight on an STM32F103 flight controller; diagnosed and resolved hardware faults on non-standard clone boards.

Self-Hosted Homelab Infrastructure — Built and maintain a self-hosted Linux NAS running containerized services (Home Assistant, local storage, media streaming) with Docker.

Workflow Automation & DevTools — Developed Python and Playwright scrapers and automation tools using the Chrome DevTools Protocol (CDP) for event management systems.

Professional Experience

Database & Systems Administrator — Academia Yamaha AGS 05/2024 – 06/2026

- Standardized student databases and automated reporting workflows with custom Python scripts and Excel VBA.
- Administered SAT-compliant electronic invoicing and inventory tracking systems.

Team Leader IS-Warehouse — Sensata Technologies 01/2023 – 03/2024

- Built automated Excel reporting models for real-time production tracking and schedule monitoring.
- Coordinated material logistics and stock availability against high-volume production demand.

Technical Skills

Programming: C, C++, Python, VBA

HDL & Assembly: VHDL, ARM Assembly

Embedded Systems: STM32 (Cortex-M, HAL & register-level), ESP32/ESP8266, PIC, AVR

Protocols: UART/USART, SPI, I2C, CAN

Tools & OS: KiCad, Git, Docker, Linux, STM32CubeIDE, PlatformIO, Node-RED, Playwright, Node.js

Languages

English: B1 / Intermediate (Professional Reading & Writing)

Soft Skills

Team Leadership, Technical Writing, Project Coordination, Troubleshooting